

SAT-Style Practice Test

June 2024 Edition (Original)

ANSWER KEY WITH FULL EXPLANATIONS

Section 1: Reading and Writing — Modules 1 & 2 (54 Questions)

Section 2: Mathematics — Modules 1 & 2 (44 Questions)

Total: 98 Questions

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Quick-Reference Answer Key

Section 1, Module 1 — Reading & Writing (27 Questions)

Q#	Ans	Q#	Ans	Q#	Ans	Q#	Ans	Q#	Ans	Q#	Ans	Q#	Ans	Q#	Ans	Q#	Ans
1	B	2	A	3	A	4	B	5	C	6	A	7	C	8	A	9	C
10	D	11	A	12	D	13	A	14	D	15	D	16	C	17	B	18	B
19	C	20	A	21	C	22	B	23	A	24	A	25	B	26	A	27	B

Section 1, Module 2 — Reading & Writing (27 Questions)

Q#	Ans	Q#	Ans	Q#	Ans	Q#	Ans	Q#	Ans	Q#	Ans	Q#	Ans	Q#	Ans	Q#	Ans
1	D	2	C	3	D	4	B	5	C	6	B	7	D	8	C	9	A
10	B	11	C	12	C	13	C	14	D	15	D	16	C	17	C	18	B
19	B	20	D	21	B	22	B	23	A	24	C	25	B	26	A	27	B

Section 2, Module 1 — Math (22 Questions)

[illegible]

Section 2, Module 2 — Math (22 Questions)

[illegible]

SECTION 1, MODULE 1: Reading & Writing (27 Questions)

Question 1 (Vocabulary in Context)

Answer: B) Throw

In the passage, Tommy throws a stone "so far that it becomes a tiny dark speck before vanishing beneath the grey waves." The word "hurl" means to throw something with great force, which matches the description of the stone traveling a great distance.

Question 2 (Vocabulary in Context)

Answer: A) echo

Lin's memorial is described as reflecting the tradition of Japanese simplicity—anyone familiar with Japanese gardens "would recognize the influence." An "echo" is a reflection or reverberation of something, making it the best fit for a design that mirrors another tradition.

Question 3 (Vocabulary in Context)

Answer: A) alleviate

The Singapore Exchange ensures transparency standards, which can reduce or ease investors' concerns about regulatory gaps. "Alleviate" means to make something less severe or to relieve, which fits the context of reducing concerns.

Question 4 (Vocabulary in Context)

Answer: B) overlook

The passage says classifying musicians into rigid genres causes critics to miss subtleties. "Overlook" means to fail to notice or to miss, which matches "may inadvertently miss nuances" in the following sentence.

Question 5 (Main Purpose)

Answer: C) To portray Ifemelu's strong commitment to collecting responses for the survey

The text shows Ifemelu guarding the survey "like a prized heirloom," carrying it at all times, approaching neighbors everywhere—at mailboxes, elevators, parking garages, and laundry rooms. This persistent behavior portrays her strong commitment to collecting survey responses.

Question 6 (Text Structure and Purpose)

Answer: A) It provides a general definition of the phenomenon discussed in the text.

The underlined sentence defines reduplication as a phenomenon "in which an element of a root word is repeated, sometimes with modification, within another word that is related to the root word." This is a general definition of the linguistic concept being discussed.

Question 7 (Reading Comprehension)

Answer: C) They have a much stronger effect on water flow than nanoplastics alone do.

The text explicitly states that emulsifiers "are released into the water from nanoplastics and have a much stronger flow-reducing effect." This directly answers what Yamamoto and colleagues discovered about emulsifiers.

Question 8 (Main Idea)

Answer: A) Continued research and development in synthetic photovoltaic technology is needed before the devices can be widely used as an energy source.

The text states the technology "is not yet commercially feasible as a large-scale power source" and describes ongoing engineering efforts to address challenges. This supports the main idea that continued R&D is needed before widespread use.

Question 9 (Data Interpretation)

Answer: C) 2,019 square kilometers.

The table shows Nisga'a Lands in British Columbia has an area of 2,019 square kilometers. The student is comparing this to his own territory's area of 39,000 square kilometers.

Question 10 (Data Interpretation)

Answer: D) Italy reported 87 damaging insect species, which is more than either Portugal or Greece reported.

From the table: Italy reported 87 insect species, Portugal reported 22, and Greece reported 7. Italy's count (87) is indeed greater than both Portugal (22) and Greece (7), directly supporting the researchers' conclusion.

Question 11 (Command of Evidence)

Answer: A) The study by Chen and colleagues had a higher alignment rating than the study by Schmidt and colleagues did.

The conclusion states that Chen's data would need less adjustment to fit the database classifications than Schmidt's data. A higher classification alignment rating directly supports this, as higher alignment means less adjustment is needed.

Question 12 (Data Interpretation)

Answer: D) The supply of fresh vegetables was ranked higher for local residents than were the other four benefits.

Local residents ranked supply of fresh vegetables at 3 (out of 20), while the other listed benefits received rankings of 10, 18, 14, and 16. A lower rank number means a higher perceived contribution, so rank 3 is the highest among the listed benefits for local residents, supporting the official's conclusion.

Question 13 (Command of Evidence)

Answer: A) None of the species that produced light did so before the simulated encounters, and nearly all stopped producing light within ten seconds after the encounters.

If the light is directed at predators (the claim), we would expect it to occur only during predator encounters and stop afterward. Choice A shows exactly this: light was produced only during encounters and stopped shortly after, supporting the claim that the signals are anti-predator responses rather than intraspecies communication.

Question 14 (Logical Completion)

Answer: D) there are geographic differences in primary language within South Asia that are reflected in the settlement patterns of South Asian immigrants in London.

The passage shows that immigrants from northern India (Hindi speakers) settled in Southall, while those from the Sylhet region of Bangladesh (Sylheti/Bengali speakers) settled in Tower Hamlets. This parallel between geographic origin and settlement patterns is best explained by geographic language differences being reflected in London settlement patterns.

Question 15 (Standard English Conventions)

Answer: D) to appear

"Making the pioneering nurse one of many healthcare professionals to appear on public health documents" is grammatically correct. The infinitive "to appear" modifies "professionals," describing what they did. The other choices create verb tense or agreement errors in context.

Question 16 (Standard English Conventions)

Answer: C) route, which is 6 miles long,

A nonrestrictive (nonessential) clause like "which is 6 miles long" provides additional information and must be set off by commas on both sides. Choice C correctly places commas before and after the clause.

Question 17 *(Standard English Conventions)*

Answer: B) spanning

"Montana, spanning 4,506 feet in elevation, reaches its highest point..." requires a participial phrase. The main verb of the sentence is "reaches." The participle "spanning" correctly modifies "Montana" without creating a run-on sentence or comma splice.

Question 18 *(Standard English Conventions)*

Answer: B) nation—Norway.

The dash before "Norway" correctly introduces the appositive that identifies the specific IMF member nation. At the end of a sentence, a single dash is the appropriate punctuation for this dramatic identification.

Question 19 *(Standard English Conventions)*

Answer: C) that,

"Conservation efforts that, through habitat restoration and scientific research, aim to bring tigers back..." The relative clause "that... aim to bring tigers back" is interrupted by the parenthetical phrase "through habitat restoration and scientific research," which needs commas around it. Choice C provides the first comma.

Question 20 *(Standard English Conventions)*

Answer: A) are

The subject is compound: "The star graph and the cycle graph." A compound subject joined by "and" takes a plural verb. Therefore "are" is correct.

Question 21 *(Standard English Conventions)*

Answer: C) dendrochronologist

The phrase "claims dendrochronologist Valerie Trouet" is an inverted attribution (like "says Dr. Smith"). No punctuation is needed between the job title and the name. Adding commas or dashes would incorrectly separate the title from the name in this attribution structure.

Question 22 *(Transitions)*

Answer: B) consequently,

The Alvin can reach depths exceeding 6,000 meters, and most hydrothermal vents are at extreme depths. As a RESULT (consequence), the Alvin can provide more data about these vents. "Consequently" correctly signals this cause-and-effect relationship.

Question 23 *(Transitions)*

Answer: A) Of course,

The passage notes that confetti was historically made from ledger rolls, then says these rolls had been replaced by computers by 2012. "Of course" signals that this transition to computers is an obvious, expected development that the reader would naturally anticipate.

Question 24 *(Rhetorical Synthesis)*

Answer: A) One way to distinguish the crested gecko from the mourning gecko is to look at their tails...

The goal is to emphasize a difference between the two geckos. Choice A directly contrasts a specific physical feature (the tails) between the two species, effectively highlighting a distinguishing characteristic.

Question 25 *(Rhetorical Synthesis)*

Answer: B) The two calendars handle leap years differently; for example, the Julian calendar adds a leap day every four years without exception, while the Gregorian calendar introduced additional rules...

The goal is to emphasize a difference. Choice B clearly identifies how the calendars differ in their leap year rules, providing specific details about the Julian approach (every 4 years, no exceptions) versus the Gregorian modification.

Question 26 *(Rhetorical Synthesis)*

Answer: A) Based on their Beaufort numbers, a storm (number 10) involves stronger winds than a near gale (number 7), and a near gale involves stronger winds than a gentle breeze (number 3).

The goal is to compare the intensity of three wind categories. Choice A systematically compares all three categories by their Beaufort numbers, establishing a clear ordering: storm > near gale > gentle breeze.

Question 27 *(Rhetorical Synthesis)*

Answer: B) It was in September 1962, in the agricultural town of Delano, California, that Dolores Huerta co-founded... the National Farmworkers Association.

The goal is to begin a narration of Huerta's co-founding of the National Farmworkers Association. Choice B uses vivid, narrative-style language ("It was in September 1962, in the agricultural town of Delano...") to set the scene, making it an effective opening for a narration.

SECTION 1, MODULE 2: Reading & Writing (27 Questions)

Question 1 (*Words in Context*)

Answer: D) nullify

Transportation costs could cancel out or eliminate the financial benefits of remittances for families in remote villages. "Nullify" means to make something void or ineffective, which precisely describes how costs could negate the benefits.

Question 2 (*Words in Context*)

Answer: C) defunct

The art installations require software that is "largely unsupported since 2017" (specifically Java applets). "Defunct" means no longer existing or functioning, which accurately describes software that is no longer supported.

Question 3 (*Words in Context*)

Answer: D) stymie

Lowering the tax threshold could hinder economic growth by "increasing the administrative burden" during a critical scaling period. "Stymie" means to prevent or hinder progress, which matches the idea of administrative burdens obstructing entrepreneurial growth.

Question 4 (*Words in Context*)

Answer: B) engendering

The surge in vinyl record values had "the counterintuitive effect of _____ demand"—new collectors flooded the market. "Engendering" means causing or giving rise to something. The price surge engendered (created/generated) new demand.

Question 5 (*Text Structure*)

Answer: C) It presents the result of a study, then raises a potential concern related to that result.

The text first presents Yamamoto's conclusion (ocean temperatures affect tuna migration), then raises the concern from Mendez that the small sample size (30 fish) could lead to biased analysis and exaggerated effects.

Question 6 (*Text Structure and Purpose*)

Answer: B) It emphasizes the potential utility of the team's model.

The underlined sentence says the model "is sure to be invaluable in determining the optimal mix of a district's businesses." This directly emphasizes the potential usefulness (utility) of the research team's predictive model.

Question 7 (*Main Purpose*)

Answer: D) To situate Tanaka and Okonkwo's study in the context of their field

The text describes prior work by Alvarez (shared pathways) and Sharma (different pathways), notes the gap in knowledge, then explains how Tanaka and Okonkwo's study was motivated to address this gap. This situates their work within the broader field of evolutionary biology research.

Question 8 (*Cross-Text Connections*)

Answer: C) At present, it only partially captures the stylistic dimensions of the fiction written in the period.

Text 1 argues that scholarship overemphasizes confessional modes while ignoring works like Spark's. Text 2 argues that scholars have focused too narrowly on "confessional sincerity versus ironic detachment," missing

other stylistic dimensions. Both authors agree that current scholarship only partially captures the full picture.

Question 9 (Main Idea)

Answer: A) A group of economists asserts that people with strong community ties are consequently motivated to take certain considerations into account when voting.

Lindqvist et al. propose that community solidarity motivates lower-income voters to support programs funded by taxes on themselves. The key insight is that social bonds (community ties) influence voting behavior, causing people to take communal welfare into account when voting.

Question 10 (Data Interpretation)

Answer: B) the wolf habitat in Scandinavia cannot support as many individuals as can the wolf habitat in Abruzzo.

The maximum density reported by Liberg (Scandinavia) was 2.10, compared to Ciucci's (Abruzzo) maximum of 7.80. This difference suggests Scandinavia's habitat cannot support as many wolves per 1,000 km² as Abruzzo can.

Question 11 (Command of Evidence)

Answer: C) Lily was immensely interested in people who possessed great fortunes. Otherwise not at all.

The claim is that Lily "admires wealth in other people to the exclusion of all other qualities." Choice C shows this perfectly: she was interested in people with fortunes, and "otherwise not at all"—meaning wealth was the only quality that mattered to her.

Question 12 (Command of Evidence)

Answer: C) It is easier to detect low and harmless concentrations of VOCs in Carrier 58MCA filters than it is to detect high and harmful concentrations of VOCs using PIDs.

The hypothesis is that HVAC filter accumulation could serve as a detection method, eliminating the need for costly PIDs. If filters can detect even low concentrations more easily than PIDs detect high ones, this strongly supports the filters' usefulness as a replacement monitoring tool.

Question 13 (Command of Evidence)

Answer: C) Cats showed a different pattern of brain activity when smelling the familiar scent vs. the unfamiliar scent, and this difference decreased as cat age decreased.

The conclusion is that younger cats are worse at differentiating familiar from unfamiliar scents. Choice C directly supports this: as age decreases (younger cats), the difference in brain response between familiar and unfamiliar scents also decreases, meaning younger cats differentiate less effectively.

Question 14 (Command of Evidence)

Answer: D) "Mill and furnace, forge and field, / Have opened wider paths before the common hand..."

The claim is that new freedoms and possibilities have become available to the working class. Choice D describes how workplaces ("mill and furnace, forge and field") have "opened wider paths before the common hand" with "greater yields" that are "more justly earned"—directly illustrating new opportunities and freedoms for workers.

Question 15 (Logical Completion)

Answer: D) some aspect of the scientific understanding of pulsar formation or neutron star evolution may be incomplete.

The passage states that conditions during neutron star formation are believed to have been "ill-suited to producing such extreme spin rates." If some pulsars still spin extremely fast despite this, it suggests our

understanding of the formation or evolution process may be incomplete.

Question 16 (Logical Completion)

Answer: C) the hypothesis about the development of this visual defense likely does not account for all instances of the trait in squid.

The hypothesis says chromatophore displays evolved because they have lower metabolic cost than ink defense. But if species with chromatophore displays ALSO produce highly toxic ink, they haven't abandoned the ink defense. This suggests the metabolic-cost hypothesis doesn't fully explain why all species developed chromatophore displays.

Question 17 (Standard English Conventions)

Answer: C) initiative: the

A colon after "initiative" correctly introduces the specific name and description of the initiative. "Timeline of Soul Food" is an appositive that elaborates on what the "important new initiative" is.

Question 18 (Standard English Conventions)

Answer: D) linguist Nomsa Dlamini

When a title is followed by a specific name (one individual is being identified), no punctuation is needed: "linguist Nomsa Dlamini of the University of Cape Town." The title and name form a single unit.

Question 19 (Standard English Conventions)

Answer: B) centuries: the

A colon after "centuries" introduces the name and description of the mercantile alliance. The text that follows ("the Hanseatic League, a loose confederation...") identifies what the alliance was.

Question 20 (Standard English Conventions)

Answer: D) ranks

The subject is "collection" (singular), not "artworks" or "artists." "Whose extensive collection... ranks among the finest" uses the singular verb "ranks" to agree with the singular subject "collection."

Question 21 (Standard English Conventions)

Answer: B) sharks Scarface and Luna

The names "Scarface and Luna" are essential (restrictive) appositives that identify which specific sharks. No commas are needed because the names are necessary information—without them, we wouldn't know which sharks are meant. There is also no comma after "Luna" because nothing follows that requires one in this context.

Question 22 (Standard English Conventions)

Answer: B) Macharia, having

The sentence needs "Macharia" as the subject of "celebrates." The participial phrase "having referred to Kenya's thriving film scene..." is set off by commas as a nonrestrictive modifier. "Macharia, having referred... celebrates" is grammatically correct.

Question 23 (Transitions)

Answer: A) In contrast,

The passage compares an older solar park (40 MW, 2008) with a newer one (187 MW, 2020). "In contrast" correctly signals this comparison between the two different performance levels.

Question 24 (Rhetorical Synthesis)

Answer: C) A reconstructed model of Titanoboa is on display at the Smithsonian National Museum of Natural History in Washington, D.C.

The goal is to specify Titanoboa's location. Choice C directly states where a reconstructed model can be found: the Smithsonian National Museum of Natural History in Washington, D.C., precisely specifying the location.

Question 25 *(Rhetorical Synthesis)*

Answer: B) Lemon juice, with a pH of approximately 2, is highly acidic, while baking soda solution (pH 9) and bleach (pH 12.5) are both basic, with bleach being the most basic of the three.

The goal is to compare the acidity or basicity of three substances. Choice B provides specific pH values for all three substances, classifies each as acidic or basic, and ranks them relative to each other—an effective comparison.

Question 26 *(Rhetorical Synthesis)*

Answer: A) Carmen Miranda's story begins in Marco de Canaveses, Portugal, in 1909.

The goal is to begin a narrative about Carmen Miranda's life. Choice A establishes the setting (Marco de Canaveses, Portugal) and the time (1909) in a narrative style ("her story begins"), making it an effective opening for a biography.

Question 27 *(Rhetorical Synthesis)*

Answer: B) Consistent with McAllester's observation that Nakai's music "draws on Native flute traditions..." Changes incorporates a traditional cedar flute.

The goal is to connect the McAllester quotation to a specific album. Choice B directly links the quotation (about drawing on Native flute traditions) to the specific album Changes (which incorporates a traditional cedar flute), effectively connecting the critical observation to the concrete work.

SECTION 2, MODULE 1: Mathematics (22 Questions)

Question 1 (Geometry — Parallel Lines)

Answer: 36

The 144° angle and angle y are co-interior (same-side interior) angles formed by a transversal crossing parallel lines. Co-interior angles are supplementary: $144 + y = 180$, so $y = 36$.

Question 2 (Inequalities)

Answer: D) $22 \leq v \leq 28$

The cyclist's speed varied between 22 and 28 km/h, meaning v is at least 22 and at most 28. This is represented by the compound inequality $22 \leq v \leq 28$.

Question 3 (Linear Equations — Scaling)

Answer: D) 24

$56x = 8(7x) = 8(3) = 24$. Since $56 = 8 \times 7$, we multiply both sides of $7x = 3$ by 8.

Question 4 (Polynomial Operations)

Answer: A) $x^3 + 9x^2 - 5x + 36$

$(x^3 + 6x^2 - 5x) + 3(x^2 + 12) = x^3 + 6x^2 - 5x + 3x^2 + 36 = x^3 + 9x^2 - 5x + 36$.

Question 5 (Function Evaluation)

Answer: 32

$f(3) = 3^3 + 5 = 27 + 5 = 32$.

Question 6 (Solving Equations with Functions)

Answer: 1272

$g(x) = x/4 = 318$. Solving: $x = 318 \times 4 = 1,272$.

Question 7 (Linear Models / Scatterplots)

Answer: D) $y = 9.5 - 0.9x$

The data shows a negative linear trend with a y -intercept around 9–10 and a negative slope of approximately -0.9 . Among the choices, $y = 9.5 - 0.9x$ best fits the pattern. Checking: at $x = 1$, $y \approx 8.6$; at $x = 10$, $y \approx 0.5$.

Question 8 (Quadratic Functions — Vertex)

Answer: A) The toy rocket's minimum height was 3 feet above the ground.

The vertex form $f(x) = (1/16)(x - 6)^2 + 3$ has vertex at $(6, 3)$. Since the coefficient $1/16$ is positive, the parabola opens upward, making the vertex a minimum. The minimum height is 3 feet.

Question 9 (Tables and Linear Equations)

Answer: A) x : 0, 3, 6 and y : 15, 20, 25

Check with $5x - 3y = -45$: At $x = 0$: $0 - 3(15) = -45$ ✓. At $x = 3$: $15 - 3(20) = 15 - 60 = -45$ ✓. At $x = 6$: $30 - 3(25) = 30 - 75 = -45$ ✓.

Question 10 (Word Problems)

Answer: 45

Total mixture = 60 mL. Medication = 15 mL. Saline = $60 - 15 = 45$ mL.

Question 11 (Systems of Equations)

Answer: B) 0

Set equal: $-(1/5)x = (1/13)x$. Moving all terms to one side: $-(1/5)x - (1/13)x = 0$. Factor out x : $x(-1/5 - 1/13) = 0$. Since the coefficient is nonzero, $x = 0$.

Question 12 (Exponential Growth)

Answer: B) $s = 2,400(1.08)^m$

An 8% increase each month means each month's count is 1.08 times the previous month's count. Starting at 2,400: $s = 2,400(1.08)^m$.

Question 13 (Exponential Functions — Doubling)

Answer: 40

$f(t) = 5,000(2)^{t/40}$. The population doubles when $2^{t/40} = 2$, i.e., when $t/40 = 1$, so $t = 40$ minutes.

Question 14 (Systems — Infinitely Many Solutions)

Answer: C) $3/8$

For infinitely many solutions, both equations must represent the same line. The first equation has slope $3/8$, so the second equation must also have slope $3/8$.

Question 15 (Quadratic Equations)

Answer: D) 6

Divide $6x^2 - 18x - 36 = 0$ by 6: $x^2 - 3x - 6 = 0$. Therefore $x^2 - 3x = 6$.

Question 16 (Geometry — Volume and Surface Area)

Answer: 864

Side length = $\sqrt[3]{1728} = 12$ cm. Surface area = $6 \times 12^2 = 6 \times 144 = 864$ cm².

Question 17 (Word Problems — Linear Equations)

Answer: 18

Before coupon: $\$91 + \$35 = \$126$. Price per notebook: $\$126 \div 7 = \18 .

Question 18 (Quadratic Equations — Discriminant)

Answer: 15

For no real solutions: $b^2 - 4(-3)(-20) < 0 \rightarrow b^2 - 240 < 0 \rightarrow b^2 < 240 \rightarrow b < \sqrt{240} \approx 15.49$. The largest positive integer is 15.

Question 19 (Trigonometry — Right Triangles)

Answer: C) 93.0

In right triangle DEF with right angle at F: $\tan(E) = \text{opposite/adjacent} = DF/EF$. So $1/5 = 18.6/EF \rightarrow EF = 18.6 \times 5 = 93.0$.

Question 20 (Quadratic Functions — Minimum)

Answer: B) 9

$f(x) = (x - 5)^2 + 9$. The minimum of $(x - 5)^2$ is 0 (at $x = 5$). Therefore, the minimum of $f(x) = 0 + 9 = 9$.

Question 21 (Circles and Angles)

Answer: D) 21π

Point A is at $(0, 1)$ and C is at $(0, -1)$. The angle ABC from ray BA to ray BC is π radians (180°). Equivalent angles: $\pi + 2k\pi$ for integer k . Testing: $21\pi = \pi + 20\pi = \pi + 10(2\pi)$. So 21π is coterminal with π . ✓

Question 22 (Triangle Similarity)

Answer: D) Neither I nor II is sufficient.

We know $\angle J = \angle M = 45^\circ$ and $KL/NO = 20/15 = 4/3$. For SAS similarity, we need two sides that include the known angle to be proportional. But KL and NO are opposite the known angle, not adjacent to it. Statement I gives one adjacent side ratio, and Statement II gives another, but each alone provides only one proportional side pair adjacent to the angle plus one opposite—not sufficient for SAS. Neither alone is sufficient.

SECTION 2, MODULE 2: Mathematics (22 Questions)

Question 1 (*Linear Functions — Constants*)

Answer: 13

$f(x) = 6x + c$. When $x = 0$: $f(0) = 6(0) + c = c = 13$.

Question 2 (*Statistics — Comparing Medians*)

Answer: C) The medians of data sets P and Q are equal.

Data set P has 6 values; its median $= (10 + 10)/2 = 10$. Data set Q has 7 values; its median = the 4th value = 10. Both medians are 10.

Question 3 (*Statistics — Margin of Error*)

Answer: A) It is plausible that the proportion is between 0.29 and 0.41.

Confidence interval: $0.35 \pm 0.06 = (0.29, 0.41)$. It is plausible the true proportion falls within this range.

Question 4 (*Linear Inequalities*)

Answer: B) (0, 6)

The shaded region is above/left of a line with positive slope through the origin (slope ≈ 2). Testing (0, 6): the line at $x = 0$ is $y = 0$. Since $6 > 0$, point (0, 6) is in the shaded region.

Question 5 (*Quadratic Functions — Tables*)

Answer: A) $f(x) = 2x^2 + 6x + 10$

Check $f(0) = 10$ ✓ (all pass). Check $f(1) = 18$: A gives $2 + 6 + 10 = 18$ ✓, B gives $7 + 1 + 10 = 18$ ✓. Check $f(-1) = 6$: A gives $2 - 6 + 10 = 6$ ✓, B gives $7 - 1 + 10 = 16$ ✗. Only A passes all checks.

Question 6 (*Exponential Growth — Graph*)

Answer: C) 1.4 times

From the graph: at $x = 0$, $y \approx 5$ (thousand); at $x = 1$, $y \approx 7$ (thousand). Ratio: $7/5 = 1.4$. The revenue at $x = 1$ is 1.4 times the revenue at $x = 0$.

Question 7 (*Linear Functions — Slope*)

Answer: A) 5

In $C(x) = 85 - 5x$, the coefficient of x is -5 . The magnitude $|5|$ represents the rate of decrease: 5 percent per hour.

Question 8 (*Geometry — Special Right Triangles*)

Answer: B) $16\sqrt{3}$

The 30° angle is at the bottom-right, the right angle is at the bottom-left, so the top angle is 60° . The vertical side w is opposite the 30° angle. Using $\tan(30^\circ) = w/48$: $w = 48 \times \tan(30^\circ) = 48 \times (1/\sqrt{3}) = 48/\sqrt{3} = 48\sqrt{3}/3 = 16\sqrt{3}$.

Question 9 (*Proportional Relationships*)

Answer: B) 2,520

$q/r = 3,150/4,200 = 3/4$. When $r = 3,360$: $q = (3/4)(3,360) = 2,520$.

Question 10 (*Geometry — Cube Surface Area*)

Answer: 9,600

Side $= \sqrt[3]{64,000} = 40$ units. Surface area $= 6 \times 40^2 = 6 \times 1,600 = 9,600$ sq units.

Question 11 (Linear Equations — No Solution)**Answer: D) 5/2**

Distribute: $(6cx + 24)/5 = 3x - 2 \rightarrow 6cx + 24 = 15x - 10 \rightarrow (6c - 15)x = -34$. For no solution, $6c - 15 = 0 \rightarrow c = 15/6 = 5/2$. The constant $-34 \neq 0$, confirming no solution.

Question 12 (Linear Functions — x-intercept)**Answer: A) When $f(x) = 0$, the number is $-21/7$.**

$f(x) = 7x + 21$. The x-intercept is where $f(x) = 0$: $7x + 21 = 0 \rightarrow x = -3 = -21/7$. This means when the output is 0, the input (the number) is $-21/7$.

Question 13 (Rational Equations)**Answer: B) $d = 5c/(3 - 4c)$**

Starting from $3/c - 5/d = 4$: rearrange to $3/c - 4 = 5/d \rightarrow (3 - 4c)/c = 5/d$. Cross-multiply: $d(3 - 4c) = 5c \rightarrow d = 5c/(3 - 4c)$.

Question 14 (Word Problems — Systems)**Answer: 3,600**

Let a = votes against. Votes for = $6a$. Difference: $6a - a = 18,000 \rightarrow 5a = 18,000 \rightarrow a = 3,600$.

Question 15 (Function Evaluation — Quadratic)**Answer: D) 432**

$f(12) = 3(12^2) = 3(144) = 432$.

Question 16 (Exponents and Radicals)**Answer: 4/5**

$\sqrt[p^3]{p^3} = p^{3/4}$. Also $t^{5/4} = (p^{2n-1})^{5/4} = p^{5(2n-1)/4}$. Setting exponents equal: $3/4 = 5(2n - 1)/4 \rightarrow 3 = 10n - 5 \rightarrow 10n = 8 \rightarrow n = 4/5$.

Question 17 (Absolute Value Equations)**Answer: C) 0 only**

$6|x - 5| = k$ gives $|x - 5| = k/6$. An absolute value equation has exactly one solution when $k/6 = 0$ (the only solution is $x = 5$). If $k/6 > 0$, there are two solutions. If $k/6 < 0$, there are no solutions.

Question 18 (Quadratic Equations — Discriminant)**Answer: 18**

For $-5x^2 + bx - 18 = 0$ to have no real solution: $b^2 - 4(-5)(-18) < 0 \rightarrow b^2 - 360 < 0 \rightarrow b^2 < 360 \rightarrow b < \sqrt{360} \approx 18.97$. The largest positive integer is 18.

Question 19 (Exponential Functions — y-intercept)**Answer: D) Neither I nor II**

$f(0) = a(1 + 2.5^b)$. For $g(x) = a(2.5^x + k)$: $g(0) = a(1 + k)$. The y-intercept value $a(1 + k)$ is not displayed as a single constant or coefficient. For $h(x) = a(2.5)^x + m$: $h(0) = a + m$. The y-intercept is $a + m$, which again is not displayed as a single visible constant. Neither form displays the y-coordinate as a constant or coefficient.

Question 20 (Linear Equations — Word Problems)**Answer: D) $y = 95x + 85$**

Cost = \$180 for the first hour + \$95 for each additional hour ($x - 1$ additional hours). $y = 180 + 95(x - 1) = 180 + 95x - 95 = 95x + 85$.

Question 21 (Percentages — Multi-step)

Answer: 60,000

Mass of X = $3.75Y$ (375% of Y). Mass of X = $0.00625Z$ (0.625% of Z). So $Z = X/0.00625 = 3.75Y/0.00625 = 600Y$. Z is 600 times Y, which is 60,000% of Y. Therefore $q = 60,000$.

Question 22 (*Triangle Similarity*)

Answer: D) Neither I nor II is sufficient.

We know angle A = angle D = 40° and $BC/EF = 18/12 = 3/2$. BC and EF are opposite the known equal angles (not adjacent to them). For SAS similarity, we need two sides that include the known angle to be proportional. Statement I gives $AB/DE = 3/2$ (one side adjacent), and II gives $AC/DF = 3/2$ (the other side adjacent). Each alone provides only one adjacent proportional side—insufficient for SAS. We would need both together. Neither alone is sufficient.

End of Answer Key — SAT-Style Practice Test: June 2024 Edition